ADJUNCTION FORMULAS FOR THE PUSHFORWARD AND PULLBACK FUNCTORS ALONG A MORPHISM OF SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one external item (a section of the same chapter family), recorded in assembly.external_references with its canonical tag as a navigation pointer.

Allowed standard prerequisites (editorial).

- Definitions of g_!, g^{-1}, u^p and g_{p!} for a morphism of sites. Stacks Project, Modules on Sites, section on morphisms of sites (definitions of the four functors). Supplies the functors whose adjunction is being established.
- Universal property of sheafification. Stacks Project, Sites, section on sheaves on a site. Converts the presheaf Hom-set into a sheaf Hom-set, giving the second formula.

Source reference. Original reference 00VD: sources/stacks-chow/tags.txt, source lines 1143–1143. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *The functor $g_{p!}$ is a left adjoint to the functor u^p. The functor $g_!$ is a left adjoint to the functor g^{-1}. In other words the formulas*

$$\begin{aligned}\mathrm{Mor}_{PAb(\mathcal{C})}(\mathcal{F}, u^p\mathcal{G}) &= \mathrm{Mor}_{PAb(\mathcal{D})}(g_{p!}\mathcal{F}, \mathcal{G}),\\ \mathrm{Mor}_{Ab(\mathcal{C})}(\mathcal{F}, g^{-1}\mathcal{G}) &= \mathrm{Mor}_{Ab(\mathcal{D})}(g_!\mathcal{F}, \mathcal{G})\end{aligned}$$

hold bifunctorially in $\mathcal{F}$ and $\mathcal{G}$.

Proof. The second formula follows formally from the first, since if $\mathcal{F}$ and $\mathcal{G}$ are abelian sheaves then

$$\begin{aligned}\mathrm{Mor}_{Ab(\mathcal{C})}(\mathcal{F}, g^{-1}\mathcal{G}) &= \mathrm{Mor}_{PAb(\mathcal{D})}(g_{p!}\mathcal{F}, \mathcal{G})\\ &= \mathrm{Mor}_{Ab(\mathcal{D})}(g_!\mathcal{F}, \mathcal{G})\end{aligned}$$

by the universal property of sheafification.

To prove the first formula, let $\mathcal{F}$, $\mathcal{G}$ be abelian presheaves. To prove the lemma we will construct maps from the group on the left to the group on the right and omit the verification that these are mutually inverse.

Note that there is a canonical map of abelian presheaves $\mathcal{F} \to u^p g_{p!}\mathcal{F}$ which on sections over U is the natural map $\mathcal{F}(U) \to \mathrm{colim}_{u(U)\to u(U')} \mathcal{F}(U')$, see Sites, Lemma 00VD. Given a map $\alpha : g_{p!}\mathcal{F} \to \mathcal{G}$ we get $u^p\alpha : u^p g_{p!}\mathcal{F} \to u^p\mathcal{G}$. which we can precompose by the map $\mathcal{F} \to u^p g_{p!}\mathcal{F}$.

Note that there is a canonical map of abelian presheaves $g_{p!}u^p\mathcal{G} \to \mathcal{G}$ which on sections over V is the natural map $\mathrm{colim}_{V\to u(U)} \mathcal{G}(u(U)) \to \mathcal{G}(V)$. It maps a section $s \in u(U)$ in the summand corresponding to $t : V \to u(U)$ to $t^*s \in \mathcal{G}(V)$.

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Hence, given a map $\beta : \mathcal{F} \to u^p\mathcal{G}$ we get a map $g_{p!}\beta : g_{p!}\mathcal{F} \to g_{p!}u^p\mathcal{G}$ which we can postcompose with the map $g_{p!}u^p\mathcal{G} \to \mathcal{G}$ above. $\square$